Barkha Dhamija

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Education

Guru Gobind Singh Indraprastha University — Expected May 2028
Bachelor of Technology in Computer Science; CGPA: 7.78/10 — *New Delhi*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Distributed Systems, Web Development

Shah International School — March 2024
Senior Secondary (Class XII) – 79.8% | Class X – 94.5% — *New Delhi*

Experience

Full Stack Developer Intern — July 2026 – Present
Mswasth — *Gurugram*

- Working with the Ops team to build and maintain full-stack features, along with analyzing operational data to surface insights and support decision-making.
- Collaborated cross-functionally with Ops to ship three full-stack features end-to-end, using operational data insights to cut a key manual review process by 30%.

Technical Content Writer Intern — June 2026 – July 2026
BrightRays — *Noida*

- Authored 20+ technical blog articles on cloud computing and other emerging technology topics, translating complex concepts into clear, accessible content for a broad technical audience.
- Maintained a twice-weekly publishing schedule across 5+ technical topic areas, growing article readership by over 40% within two months.

Projects

DistroQ – Distributed Task Queue | *Go, Redis, PostgreSQL, gRPC, Docker* — GitHub

- Designed and built a distributed task queue in Go using a worker-pool architecture with goroutines and channel-based synchronisation, processing **>10,000 jobs/min**.
- Engineered at-least-once delivery via Redis job locking and a PostgreSQL dead-letter queue with hash-map deduplication, cutting reprocessing overhead by **60%**.

ELIRA | *Flutter, Dart, Go, Firebase, Riverpod* — **Top 30 – Eclipse 6.0, Thapar Hackathon** | GitHub

- Architected a distributed full-stack system across a Flutter mobile client and Go backend, designing Cloud Firestore schemas and async data models to handle real-time state sync across **10+ screens** without data races.
- Implemented Riverpod StateNotifier for deterministic, concurrent state management and integrated Firebase Auth (OTP) with **30-day** persistent sessions, supporting reliable access in high-stress environments.

Pixel – Photo Culling Tool | *React, TypeScript, Vite, Tailwind CSS, Zustand* — GitHub | Live

- Built a fully client-side tool for culling large event photo batches, using `react-window` to virtualize a grid of **hundreds of photos** with single-decode thumbnailing, and Zustand + IndexedDB for persistent, refresh-proof sessions.
- Designed a keyboard-driven rating workflow (pick/reject/maybe, auto-advance) with near-duplicate detection to group similar burst shots, non-destructive live edits, and `jszip`-based export of picks with configurable size and naming.

Achievements

- **Half CA** | Live – Fintechstico v7.0, Consilium'26 (NSUT): **2nd Place** in a 24-hour offline FinTech hackathon (prize pool INR 1.5L) with an AI-powered GST reconciliation assistant.
- **Top 100 Mentee – Codess Café:** selected among 2,500+ applicants; program covered CS fundamentals and interview prep.

Leadership Experience

- **Founder & Lead, ACM-W Chapter, USICT** (Jan 2026 – Present): establishing the university's first ACM-W Chapter for 200+ female students.
- **Web Development Lead, TechSpace, USICT** (Nov 2025 – Jan 2026): led end-to-end React.js development of the official TechSpace website for a 500+ member society.

Technical Skills

Languages: Go, JavaScript (ES6+), Dart, Python, C/C++, HTML5, CSS3, SQL
Frameworks & Libraries: Flutter, React.js, Next.js, Node.js, gRPC, Riverpod, Express, TailwindCSS, WebRTC
Cloud & Databases: Firebase (Auth, Firestore), Redis, PostgreSQL, MongoDB, AWS S3, Vercel, Docker
Concepts: Distributed Systems, Concurrency & Goroutines, System Design, Data Structures & Algorithms, REST/gRPC APIs, State Management, Mobile Architecture